



YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING



Department of Computer Technology

Title Selection Seminar

Advanced monitoring and control mechanism for electric vehicle charging station using Machine learning

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SECTION : A & B

YEAR: 2024-25

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COLLEGE GUIDE:

Prof.N.U.Sambhe

Department Vision and Mission

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Program Outcomes/Program Specific Outcomes

At the end of Program, Graduate students will have

- PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

Program Outcomes/Program Specific Outcomes

- PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).
- PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

Program Outcomes/Program Specific Outcomes

- PO11: Lifelong Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies
- [PSO 1] An ability to acquire skills to design & develop quality software using cutting-edge technologies of Computer Engineering.
- [PSO 2] An ability to use knowledge of various domains as per thrust areas to formulate and implement ideas for providing innovative solutions.

Course Outcomes

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	1	2	3	4	5	6	7	8	9	10	11	PSO 1	PSO 2
Identify a problem, analyze research gaps through a literature survey, and propose an ethical, innovative solution with societal impact.	3	3	3	3	3	2	2	3	3		2	3	2

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	1	2	3	4	5	6	7	8	9	10	11	PSO 1	PSO 2
Develop a structured project report detailing the problem, methodology, proposed solution, and findings, adhering to professional and ethical standards.	3	3	3	3	3	3	2	3	3		2	3	3
Implement a basic prototype/model to validate the proposed solution, demonstrating practical application of knowledge and technical skills.	3	3			3			3	3	3	3	3	3
Avg	3	3	3	3	3	2.5	2	3	3	3	2.3	3	2.6667

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Introduction

As electric vehicles (EVs) become an integral part of our daily lives, the need for smarter and more efficient charging infrastructure has never been greater. Charging stations must do more than just provide power — they need to be reliable, adaptable, and capable of meeting growing demand while minimizing energy waste.

Our project tackles this challenge by creating a predictive model using a data-driven approach and supervised machine learning. By analyzing both historical and real-time data, the model can forecast charging demand at public stations, allowing energy to be distributed efficiently, reducing grid stress, and enhancing the user experience.

But it doesn't stop there. The project also integrates advanced monitoring and control mechanisms, enabling real-time fault detection and better energy management. These features ensure charging stations remain efficient, safe, and capable of adapting to fluctuating energy loads and user needs.

Importance of selecting a relevant and impactful title in current scenario

The Title, “**Advanced monitoring and control mechanism for electric vehicle charging station using Machine learning**” perfectly captures the heart of the project by focusing on real-world challenges in EV charging.

- **Solves Real Problems:** Predicting charging demand helps ensure enough energy is available, reduces waiting times, and prevents grid overload.
- **Supports Smarter Systems:** It strengthens LVDC microgrids, making them more reliable and adaptable to changing energy needs.
- **Matches Global Trends:** Fits the push for renewable energy, smart grids, and EV-friendly systems around the world.
- **Future-Ready:** Uses advanced machine learning to keep up with the growing EV demand and make charging infrastructure better for everyone.

Literature Review

Ref no, Year of publica tion	Title	Methodology Used	Dataset Used/Existing real time application	Performance parameters used	Performance Analysis	Limitations /Gaps
[1],2020	Data-Driven Charging Demand Prediction at Public Charging Stations Using Supervised Machine Learning Regression Methods	Gradient Boosting (XGBoost), random Forest (RF), Support Vector Machine (SVM)	The research utilized seven years of charging data from Nebraska to explore applications in scheduling, grid stability, and smart grids.	Coefficient of Determination Root Mean Square Error (RMSE), Mean Absolute Error (MAE)	XGBoost achieved an RMSE of 6.68 kWh and an R-squared of 51.9%.	Limited dataset, potential difficulties in real-world application, and unforeseen challenges.
[2],2024	Automatic Fault Detection And Analysis Of Electric Vehicle Charging Station By Machine learning	SVM, tree-based models, neural networks, semi-supervised learning, XGBoost.	Real-world EV data (charging rate, temperature, SOC, mileage), load resistance, simulated fault datasets.	Fault detection accuracy, training/executio n time, algorithm generalizability, statistical error distribution, precision, recall.	XGBoost achieved 90.34% accuracy; proposed ML algo. reached 94.59% accuracy;	Inefficient traditional methods, limited real-world datasets, lack of scalability, need for higher accuracy/ speed.

Ref no, Year of publication	Title	Methodology Used	Dataset Used/Existing real time application	Performance parameters used	Performance Analysis	Limitations /Gaps
[3],2024	Data-driven Short-term Electric Vehicle Charging Station Demand Forecasting	Random Forest,Support Vector Machine,Reinforcement Learning	-Battery Performance Data, Charging and Discharging Cycles Data. -Charging Stations, Battery Management Systems.	Mean Absolute Error,Root Mean Square Error, Coefficient of Determination (R^2),Mean Absolute Percentage Error	High prediction accuracy achieved but requires optimization for real-time deployment.	Data scarcity, integration challenges, and limited adaptability to dynamic conditions.
[4],2023	Machine Learning-based Electric Vehicle User Behavior Prediction	supervised learning techniques, feature engineering, data cleaning, and preprocessing.	Public charging dataset from Caltech Adaptive Charging Network (ACN), weather data from local station, and U.S. federal holidays calendar (2019)	Mean Absolute Error (MAE), Symmetric Mean Absolute Percentage Error (SMAPE), Root Mean Squared Error (RMSE), R^2 score	Achieved MAE of 1.45 kWh (energy consumption) and 66.5 minutes (session duration); improved prediction errors	Highly stochastic nature of public charging behavior compared to residential; dataset limited to a single location and year.

Ref no, Year of publicat ion	Title	Methodology Used	Dataset Used/Existing real time application	Performance parameters used	Performance Analysis	Limitations /Gaps
[5],2023	Quality of Service Evaluation and Forecast for EV Charging Based on Real-World Data.	M/G/K queue model, SARIMAX forecasting, simulation framework.	6M records from Hydro-Quebec (2018-2022).	Charging Requests, Site Utilization, Occupancy, Blocking Probability, Waiting EVs, Avg. Waiting Time, QoS Evaluation.	QoS evaluation, demand forecasting, simulation of charging behavior.	Data gaps, regional generalization, COVID data gaps, limited exogenous factors, scalability issues.
[6],2023	Optimizing Electric Vehicle Charging Station Placement in Urban Areas: A Data-Driven Approach.	Data Analysis, ML(k-means clustering,),Optimization Algorithm.	Real-world driving and charging data from metropolitan urban areas.	Peak-to-Average Ratio (PAR) Charging Waiting Time Operating Cost.	Reduced waiting time (67.6 to 22.8 min), power consumption (PAR 3.17 to 1.23), and costs (electricity 17.4%, maintenance 31.8%)	Limited dataset and user behavior variability Missing real-time traffic/weather data, generalization issues.

Ref no, Year of publicat ion	Title	Methodology Used	Dataset Used/Existing real time application	Performance parameters used	Performanc e Analysis	Limitations /Gaps
[7],2023	Enhancement of Charging Efficiency of Batteries for Electric Vehicles: Review.	Support Vector Machines,Fuzzy Logic Systems,Artificial Neural Networks (ANN).	Datasets: Battery performance, charge cycles. Applications: EV stations, BMS.	Charging efficiency, state of charge (SOC) prediction accuracy, battery health monitoring effectiveness.	Machine learning improves SOC prediction accuracy, advanced BMS enhances battery monitoring and protection.	High Computational Demands,Data Availability,affordable charging challenges.
[8],2023	Design and Development of Smart Charging Infrastructure for Electric Vehicles.	Cloud-based intelligent models, , Linear and Binary Search for station filtering, priority scheduling for slot allocation..	Combined dataset of 24,740 US EV stations and 3,395 charging sessions from Harvard Dataverse.	Accuracy , classification metrics (confusion matrix), price and distance prediction accuracy, scheduling efficiency.	Gradient Boosting showed high accuracy,, effective station filtering with search algorithms.	Limited real-time dynamic factors, less personalized user preferences, V2G functionality not implemented.

Ref no, Year of publicati on	Title	Methodology Used	Dataset Used/Existing real time application	Performance parameters used	Performance Analysis	Limitations /Gaps
[9],2022	Electric Vehicle Charging Behavior Prediction using Machine Learning Models	K-means clustering,Machin e Learning models (Random Forest, XGBoost).	Adaptive Charging Network (ACN) dataset with over 30,000 charging sessions from Pasadena, California.	Silhouette Index for clustering; Mean Absolute Percentage Error (MAPE) for prediction	K-means clustering (0.7726–0.7879). XGBoost achieved a MAPE of 0.146%, outperforming Random Forest (0.151%)	Limited to charging sessions from a specific location; cannot generalize findings globally, exploration needed for scalability and variations in real-world conditions.
[10],2022	Machine Learning for Solving Charging Infrastructure Planning: A Comprehensive Review	Reinforcement Learning (RL), Linear Regression Mode, Support Vector Machine,Deep Learning	Real-world applications in Helsinki and Dundee for hotspot detection and grid planning.	Accuracy, error rates, costs, battery capacity, and efficiency	Supervised for accuracy; reinforcement for management	Data scarcity, integration issues, limited adaptability, and V2G gaps.

Ref no, Year of publicat ion	Title	Methodology Used	Dataset Used/Existin g real time application	Performance parameters used	Performance Analysis	Limitations /Gaps
[11],2023	Data-Driven Analysis of EV Energy Prediction and Planning of EV Charging Infrastructure.	SARIMAX, Meta Fusion Regression, K-Means Clustering Algorithm.	California EV Charging Station Dataset Palo Alto Charging Station Dataset.	Mean Absolute Error (MAE), Root Mean Square Error, Coefficient of Determination.	SARIMAX: $R^2 = 0.97$, MAE = 6.50, RMSE = 17.43 Meta Fusion Regression: $R^2 = 0.98$, MAE = 0.91, RMSE = 2.26	Limited dataset and user behavior variability, missing real-time traffic/weather data, generalization issues, and need for more resources.
[12],2021	Analysis of Energy Consumption at slow Charging Infrastructure for Electric Vehicle.	Statistical analysis, transformed beta distribution, machine learning.	Geospatial datasets, historical electricity data, EV infrastructure planning.	Energy consumption distribution, location features, charging events, statistical significance.	High income/expensive housing increases usage, social assistance areas decrease it, XGBoost performs well.	Limited inclusion of location- specific features, insufficient focus on dynamic spatial and temporal aspects, reliance on deterministic models.

Ref no, Year of publicati on	Title	Methodology Used	Dataset Used/Existing real time application	Performance parameters used	Performance Analysis	Limitations /Gaps
[13],2020	Machine Learning Approaches for EV Charging Behavior: A Review.	Decision trees,SVMs,neur al networks, clustering, and regression models.	EV charging station logs,user behaviour patterns and simulation data.	Accuracy, precision, recall, and latency.	Decision trees,SVMs, and neural networks can effectively predict and optimize charging behaviour.	Limited diversity in datasets, lacks of real-world implementation and scalability testing in varied geographic regions.
[14],2021	Analysis of Energy Consumption at Public Charging station.	Data-centric analysis, statistical methods.	Geospatial and socioeconomic datasets; charging events.	Energy modeled by transformed beta distribution; key factors include transactions, income, housing proximity, and social assistance.	Higher consumption linked to prosperous areas and businesses; lower in areas with high social assistance.	Time-intensive data collection; needs refined prediction models for deployment and grid planning.

Patent Search

Title	Publication Year	Publication Type	Abstract
Vehicle power source device and its method for diagnosing faults	2018	China Patent	The invention provides a vehicle power source device with fault diagnosis capabilities to simplify identifying issues in electrical paths during charging and power delivery.
Power grid aware machine learning device	2022	U.S Patent	A system uses machine learning to optimize the operation of electrical devices by forecasting power source status and generating operational rules, enhancing energy efficiency, grid stability, and cost savings.

Gaps Identified

1. **Session-Level Demand Prediction:** Limited studies focus on predicting energy demand at the individual session level, which is crucial for real-time management.
2. **Dynamic Charging Behavior Analysis:** Insufficient exploration of how variations in user behavior, charging locations, and session timing impact energy demand.
3. **Integration with Grid Management:** Existing solutions lack a comprehensive approach to integrate charging demand predictions with grid reliability and optimization strategies.
4. **Limited Dataset Utilization:** Many studies use small or geographically constrained datasets, limiting the generalizability of findings to diverse charging scenarios.
5. **Scalability for Rapid EV Growth:** Existing predictive models struggle to scale with the rapid growth of EVs and the increasing complexity of charging demand patterns

Problem Statement

1. **Predicting Charger Current Output:** The need for a machine learning model to predict the optimal current output at EV charging stations, ensuring efficient charging.
2. **Fault Detection and Correction:** Addressing the challenge of real-time fault detection in charging equipment, with an automated control mechanism to correct faults or alert authorized personnel if correction is not possible.
3. **Optimizing Charging Operations:** Enhancing charging station performance by dynamically optimizing charging parameters to prevent load imbalances, reduce downtime, and ensure a seamless and efficient user experience.

How the Proposed Research Fills the Gaps

1. **Lack of Real-Time Charger Prediction:** Current EV chargers lack smart systems to predict and adjust charging currents efficiently during operation; this research introduces ML-based forecasting.
2. **Missing Fault Detection Mechanisms:** Existing chargers rarely detect faults in real-time; this study uses ML to monitor faults, enabling quick fixes or instant alerts.
3. **Static and Inefficient Operations:** Many chargers lack dynamic optimization, causing inefficiencies. This research applies ML to enhance real-time performance, reduce downtime, and boost user satisfaction.

Objectives

Primary Objective/Problem Definition:

To design and develop a data-driven predictive model using supervised machine learning regression methods to accurately forecast charging demand at public charging stations, and to propose advanced monitoring and control mechanisms to optimize energy management, reduce peak demand, and enhance the overall charging experience.

Specific Objectives:

- 1.To Collect Dataset:** Gather historical data on electric vehicle charging patterns, energy consumption, and other relevant factors to support predictive modeling.
- 2.To design and develop model for Predictive analysis:**Develop a machine learning-based predictive model that utilizes collected data to forecast charging demand and energy consumption.
- 3.To monitor and Predict:** Design a real-time monitoring system that utilizes predictive analytics to forecast charging demand, detect anomalies, and optimize charging station operations.
- 4.To detect and manage fault:** Develop a fault detection and management system that uses automated control mechanism to identify potential faults, alert operators, and provide recommendations for maintenance and repair.

Rationale for Proposed Title

- **Domain:**

- IOT/Product Development
- Multidisciplinary: Yes

- **Why This Title?**

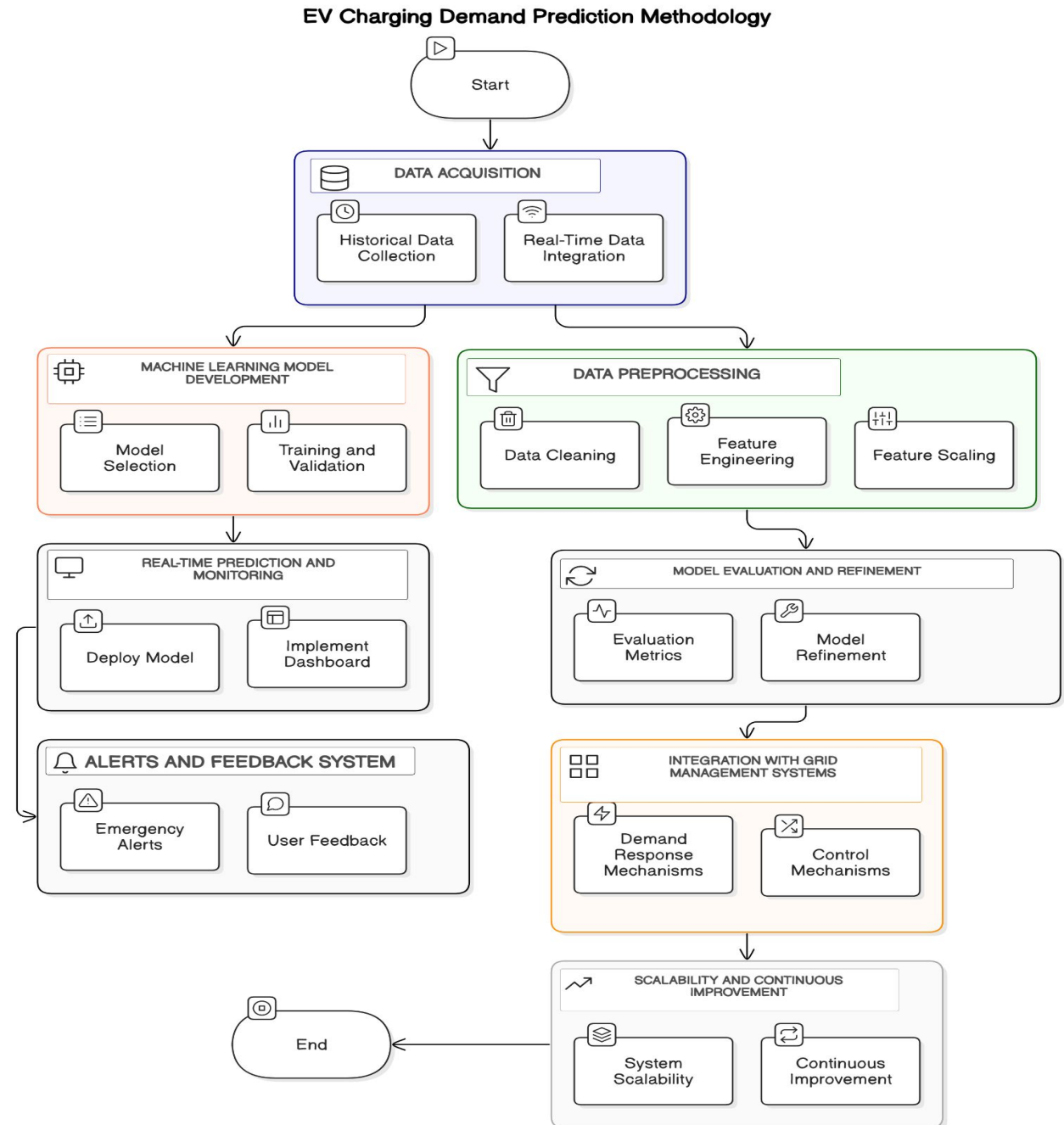
The title encapsulates the primary challenge: managing and optimizing smart charging systems to ensure safety, efficiency, and real-time control.

- **Alignment with Research Objectives:**

- The title represents the goal of the project: to create a smart charging station solution that utilizes real-time monitoring, predictive analysis, and optimization.

Methodology Overview

1. Data Acquisition
2. Data Preprocessing
3. Model Development
4. Model Evaluation
5. Real-Time Prediction
6. Grid Integration
7. Alarm and Feedback System



Expected Outcomes

Outcome 1: A machine learning model that dynamically adjust voltage/current according charging demand and energy consumption for **faster charging** while maintaining battery health.

Outcome 2: Real-time **Fault Detection and Management** and sends alerts for voltage/current anomalies to prevent hardware failures.

Outcome 3: **Optimized grid** load by managing power distribution during peak hours.

- Patent: Yes/NO (*To be decided by Committee and Guide)
- Research Paper: Conference/Journal (*To be decided by Committee and Guide)

(*Note: POST - PROJECT SUBMISSION REQUIREMENT: : As mentioned earlier, the project titles and the scope of projects shall be such that at least ONE SCOPUS / WoS indexed publication / Book Chapter / Patent is derived out of the project work. The student has to pay the fees for publication. It shall clarify to all Project Guides and the project students that if the desired aim is not achieved before ESE Viva Voce of VII Sem ESE, the project may be extended to next SIX months and the final viva voce of project shall be conducted at the time of VIII Sem ESE.)

Feasibility and Timeline

Software Tools:

1. Backend Technologies (Django)

- **Django**: For building the backend.
- **Django REST Framework (DRF)**: To create RESTful APIs
- **Django Channels**: To handle WebSocket connections for real-time updates.

2. Frontend Technologies

- **Django Templates**: For server-side rendering of HTML pages.

3.Database

- **MySQL**: For structured data like logs, predictions, and user information.
- **MongoDB**: (Optional) For storing unstructured or real-time log data.

Feasibility and Timeline

4. Machine Learning Technologies

- **Scikit-learn**: For building regression, classification, or clustering models.
- **Pandas**: For data manipulation and preprocessing.
- **NumPy**: For numerical computations.
- **Matplotlib** and **Seaborn**: For static data visualization.
- **Jupyter Notebook**: For developing, testing, and analyzing machine learning models.
- **Tensorflow** : Used for building and deploying machine learning and deep learning models.

5. Hardware Integration Technologies And Tools

Microcontrollers

- **Arduino**: For collecting analog/digital data from voltage, current, or temperature sensors.
- **ESP32/ESP8266**: For IoT capabilities with built-in Wi-Fi.
- **Raspberry Pi**: For edge computing and acting as a bridge between hardware and backend.

Django Channels: To handle WebSocket connections for real-time updates.

Feasibility and Timeline

6.Datasets:

Voltage Dataset: This dataset contains records of voltage readings across different charging stations, measured over time

Current Dataset : This dataset records the current drawn by vehicles at various charging stations, measured over time

7.Machine Learning Model

i. Predict Energy Consumption

Objective: Predict the energy consumption of vehicles

Linear Regression: If the relationship between features (voltage, current, duration) and target (energy consumption) is linear.

Gradient Boosting Models (XGBoost, LightGBM): For high-performance regression tasks with faster computations.

Feasibility and Timeline

ii. Predictive Maintenance

Objective: Predict hardware failures in the charging station to ensure seamless operation.

Logistic Regression: For binary classification (e.g., "Fail" vs. "No Fail").

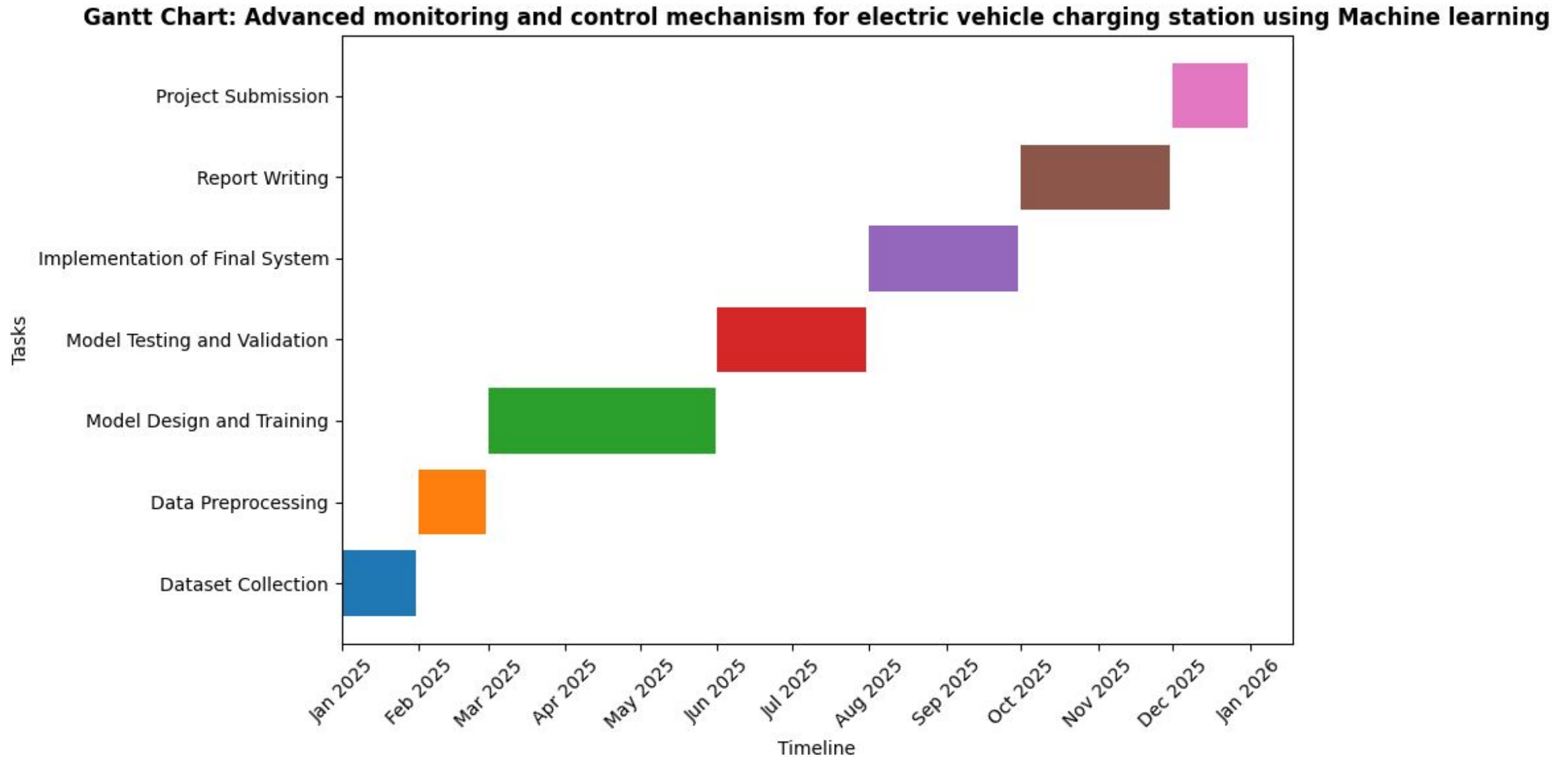
Random Forest Classifier: Handles imbalanced datasets and complex relationships well.

iii. Optimize Voltage and Current Delivery

Objective: Determine the best voltage/current settings for efficient and safe charging. and Understand and predict user behavior, such as preferred charging times or frequently visited stations.

K-Means Clustering: Groups charging scenarios based on voltage, current, and other factors.

Gantt Chart



Conclusion

Name of similar project titles from previous years:

Session	Title	Guide
-	-	-

Proposed Title Reiterated: (*To be finalized by Committee and Guide)

Summary of Justification:

- **Optimizes Energy Management:** The proposed mechanism uses machine learning algorithms to predict energy demand and adjust charging rates accordingly, reducing peak demand and energy waste.
- **Provides Reliable Charging Services:** Real-time monitoring and automated control ensure that charging stations are always available and functioning correctly, providing a reliable charging experience for electric vehicle owners.

- **Addresses a significant gap.**
- Existing systems often struggle to balance grid load and energy demand during peak hours. The ML-driven approach predicts and adjusts energy distribution dynamically.
- Many charging stations lack advanced fault detection and maintenance scheduling capabilities. ML-based monitoring enables proactive identification of issues.

References(*Strictly in IEEE format)

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Reference Summary

No of IEEE Journal	No of Conference Papers	No of Dataset	No of Web References	No of papers from last five years of publication	Paper titles with which you will show comparison
	14	-	4	-	-

Thank You